

MATH5320 – Homework IV

1)

1. Parametric VaR, 1 stock

The mean and standard deviation of the daily log returns of a stock S are 0.015% and 3.50%, respectively. Suppose the stock follows a geometric Brownian motion with parameters μ and σ , so that

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where W_t is a standard Brownian motion.

(a) What are good estimates for the drift and volatility, μ and σ , respectively?

(b) What is the 5 day 99% VaR of a position of 1400 shares in the stock if the current stock price is \$82 a share?

We assume the stock, follows a GBM, given as

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

And then, under the GBM, a one-day log return, is such that $r = \log(S_{t+1}/S_t)$, and it is normal such that:

$$E[r] = (\mu - \frac{1}{2}\sigma^2)\Delta t, \quad \text{Var}(r) = \sigma^2\Delta t$$

a)

We are given that the log-return mean is 0.015% or 0.00015 and stdev is 3.50%, 0.035, and then $\Delta t = 1$ day. From the slides:

With sample mean and variance of daily log returns being $\bar{\mu}$ and $\bar{\sigma}^2$:

$$\begin{aligned}\bar{\mu} &\approx E[\log(S_{t_2}/S_{t_1})] = \left(\mu - \frac{\sigma^2}{2}\right)(t_2 - t_1) \\ \bar{\sigma}^2 &\approx \text{var}[\log(S_{t_2}/S_{t_1})] = \sigma^2(t_2 - t_1)\end{aligned}$$

From the variance relation, we know that:

$$\sigma = 0.035 * \Delta\sqrt{(1)} = 0.035$$

From the mean-relation:

$$\mu - \frac{1}{2}\sigma^2 = 0.00015 \rightarrow \mu = 0.00015 + \frac{1}{2}\sigma^2 \rightarrow \mu = 0.00015 + \frac{1}{2}0.035^2$$

$$\mu = 0.00015 + 0.0006125 = 0.0007625 \text{ per day}$$

As a result, on a daily basis:

$$\hat{\sigma} = 0.035, \quad \hat{\mu} = 0.0007625$$

If we want the estimated terms on an annual basis, we scale accordingly, where vol scaled with $\sqrt{T}$, and mean returns scale with T , such that:

$$\begin{aligned}\widehat{\sigma}_{\text{annual}} &= 0.035 * \sqrt{252} = 0.5556 = 55.56\%, \\ \widehat{\mu}_{\text{annual}} &= 0.0007625 * 252 = 0.19215 = 19.215\%\end{aligned}$$

b)

We want to get the $T = 5$ -day, 99% VaR, so $p = 0.99$ of 1,400 shares,

Computations:

$$dS = \mu S dt + \sigma S dW$$

- $E[S_T] = S_0 e^{\mu T}$
- $\text{var}[S_T] = S_0^2 (e^{\sigma^2 T} - 1) e^{2\mu T}$
- $\text{VaR}(S, T, p) = S_0 - S_0 e^{\sigma \sqrt{T} \Phi^{-1}(1-p) + (\mu - \frac{\sigma^2}{2}) T}$
- $\text{ES}(S, T, p) = \text{homework}$
- Mostly easy to compute using martingale techniques.

Here, we note that the

$\Phi^{-1}(0.01) \approx -2.3263$, where we will use this later.

We then, continue to solve for the other parts in the exponent, where we have:

$$\sigma \sqrt{T} = 0.035 \sqrt{5} = 0.07826$$

$$0.07826 * (-2.3263) = -0.18210$$

$$\left(\mu - \frac{1}{2}\sigma^2\right)T = 0.00015 * 5 = 0.00075$$

Then, the exponent yields:

$$-0.18210 + 0.00075 = -0.18135$$

Then, the following this, 1% price quantile yields:

$$S_{1\%} = 82 e^{-0.18135} = 68.40.$$

On a per-share basis:

$$82 - 82 e^{-0.18135} = 82 - 68.40 = 13.60$$

For 1,400 shares, we get a 5day VaR of:

$$VaR_{5d,99\%} = 1400 * 13.60 = \$19,040$$

Resultingly, the 5-day 99% VaR yields, \$19,040.

*On the other hand, we can choose to use annualised numbers, $T = 5/252$, and arrive at similar numbers. Note that **rounding** might be a bit different.*

2)

2. Parametric VaR, 2 stocks

We have 400 shares of stock 1 and 600 shares of stock 2. Stock 1's current price is \$102 per share and stock 2's current price is \$81 per share. Stock 1 and 2 follow GBM with drift 3.50% and 2.30% and volatility 33% and 22%, respectively, each driven by Brownian motions with 31.00% correlation. What is the 2 week 99% VaR of the portfolio? Calculate it parametrically assuming that the portfolio is normally distributed.

Parametric VaR is utilised to make simplifying assumptions to get a VaR formula from approximate mean and variance. We are given:

- 400 shares of Stock 1, $P = 102 \rightarrow V_1 = 400 * 102 = 40,800$
- 600 shares of Stock 2, $P = 81 \rightarrow V_2 = 600 * 81 = 48,600$

As a result, the total $V_0 = 89,400$. From this, the approximate PnL, can be computed by considering:

$$\Delta V_t = V_1 R_t^{(1)} + V_2 R_t^{(2)}, \quad \text{for } 0 \leq t \leq T$$

This is true from *linearity of expectations*.

It is also given, in the question that:

- Stock 1, $\mu_1 = 3.5\%$ $\sigma_1 = 33\%$
- Stock 2, $\mu_2 = 2.3\%$ $\sigma_2 = 22\%$
- Given correlation of $\rho = 0.31$

We have:

$$S_{1,t} S_{2,t} = S_{1,0} S_{2,0} e^{(\mu_1 + \mu_2 - (\sigma_1^2 + \sigma_2^2)/2)t + \sigma_1 W_{1,t} + \sigma_2 W_{2,t}}$$

$$\sigma_1 W_{1,t} + \sigma_2 W_{2,t} \sim \mathcal{N}(0, (\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)t)$$

So, $S_1 S_2$ is lognormal and we know the variance of the Brownian motion part, so we can use our formulas for the mean of a lognormal (or just factor out the martingale part):

$$E[S_{1,t} S_{2,t}] = S_{1,0} S_{2,0} e^{(\mu_1 + \mu_2 + \rho\sigma_1\sigma_2)t}$$

$$E[V_t] = a S_{1,0} e^{\mu_1 t} + b S_{2,0} e^{\mu_2 t}$$

$$E[V_t^2] = a^2 S_{1,0}^2 e^{(2\mu_1 + \sigma_1^2)t} + b^2 S_{2,0}^2 e^{(2\mu_2 + \sigma_2^2)t} + 2ab S_{1,0} S_{2,0} e^{(\mu_1 + \mu_2 + \rho\sigma_1\sigma_2)t}$$

$$\text{var}[V_t] = E[V_t^2] - E[V_t]^2$$

From here, we are given that we have two weeks, such that $T = 10/252$ years. And we want the 99% VaR, or the 1% left tail such that we have:

$$\Phi^{-1}(0.01) = -2.32$$

Moving on, we, consider the expectation of the portfolio based off the drifts such that we have:

$$E[\Delta V] = V_1 R_1 T + V_2 R_2 T, \text{ where we have:}$$

- $V_1 \mu_1 = 40,800 * 0.035 = 1,428$
- $V_2 \mu_2 = 48,600 * 0.023 = 1,117.8$

Then, the total of 2,545.8, we take the two-week basis of $T = 10/252$

$$E[\Delta V] = 2,545.8 * \left(\frac{10}{252}\right) = 101.02380952380952$$

Now, the variance needs to be computed, and for two random mvariables,

For any two random variables, we know that:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Then, we have the following:

$$\text{Var}(\Delta V) = V_1^2 \text{Var}(R_1) + V_2^2 \text{Var}(R_2) + 2V_1 V_2 \text{Cov}(R_1, R_2).$$

Note that we are emulating this calculation from slide 25:

At a horizon t :

$$\begin{aligned} E[V_t] &= aE[S_{1,t}] + bE[S_{2,t}] \\ E[V_t^2] &= E[a^2 S_{1,t}^2 + b^2 S_{2,t}^2 + 2ab S_{1,t} S_{2,t}] \\ \text{var}[V_t] &= E[V_t^2] - E[V_t]^2 \\ \text{sd}[V_t] &= \sqrt{\text{var}[V_t]} \end{aligned}$$

Know $E[S_{1,t}^2]$ and $E[S_{2,t}^2]$. Just need $E[S_{1,t} S_{2,t}]$.

And because, we know the following for a GBM, which is normal, over T , we have:

- $\text{Var}(R_i) \approx \sigma_i^2 T$
- $\text{Cov}(R_1, R_2) = \rho \sigma_1 \sigma_2 T$
- $\text{Var}(\Delta V) = T(V_1^2 \sigma_1^2 + V_2^2 \sigma_2^2 + 2V_1 V_2 \rho \sigma_1 \sigma_2).$

Then, plugging in the values that were given, ie:

$$V_1 = 40,800, V_2 = 48,600, \sigma_1 = 0.33, \sigma_2 = 0.22, \rho = 0.31, T = \frac{10}{252}.$$

Ergo, the first term will be:

$$V_1^2 \sigma_1^2 = (40,800)^2 (0.33)^2 = 181,679,296$$

$$V_2^2 \sigma_2^2 = (48,600)^2 (0.22)^2 = 114,518,864$$

$$2V_1V_2\rho\sigma_1\sigma_2 = 2(40,800)(48,600)(0.31)(0.33)(0.22) = 89,253,394.56$$

As a result,

$$\begin{aligned}\text{Var}(\Delta V) &= \frac{10}{252} (181,679,296 + 114,518,864 + 89,253,394.56) = \frac{10}{252} (385,079,556) \\ &= 15,271,887.0857\end{aligned}$$

Resultantly, the std, gets:

$$sd(\Delta V) = \sqrt{\text{Var}(\Delta V)} = \sqrt{15,281,727} = 3,907.926187 \text{ (nearest integer)}$$

Then, assuming a normal distribution, which is the key idea of a parametric VaR, and assuming linear payoffs, we have:

$$\Delta V \sim N(m, s^2), \text{ with}$$

$$m = E[\Delta V] = 101.02(2dp), \quad s = sd(\Delta V) = 3,907.93$$

As such, the 1% quantile is:

$$m + \Phi^{-1}(0.01)s = 101.02 + (-2.3263)(3,908) = -8,990.172$$

As a result, the 2-week 99% VaR, is:

$$VaR_{0.99} = 8,990.172$$

Note that this is methodologically similar to Slide 24 - 30 calculations, accounting for V_0 , and expectations of the V_t , but it is much cleaner to look purely at PnLs.

Using the same way as the slides, we have once again, that the:

$$E[V_T] = V_0 + E[\Delta V] \approx 89,400 + 101.02 = 89,501.02$$

$$sd(V_T) = 3,908$$

From here, we use the following rule:

Normal distribution:

- The 1% tail is 2.326 standard deviations out.

Assuming V_t is normal:

$$VaR[V] = V_0 - (E[V_t] - 2.326sd[V_t])$$

Therefore,

$$V_{T,1\%} = E[V_T] - 2.326sd(V_T)$$

$$VaR_{99\%} = V_0 - V_{T,1\%} = 8,990 \text{ (approx)}$$

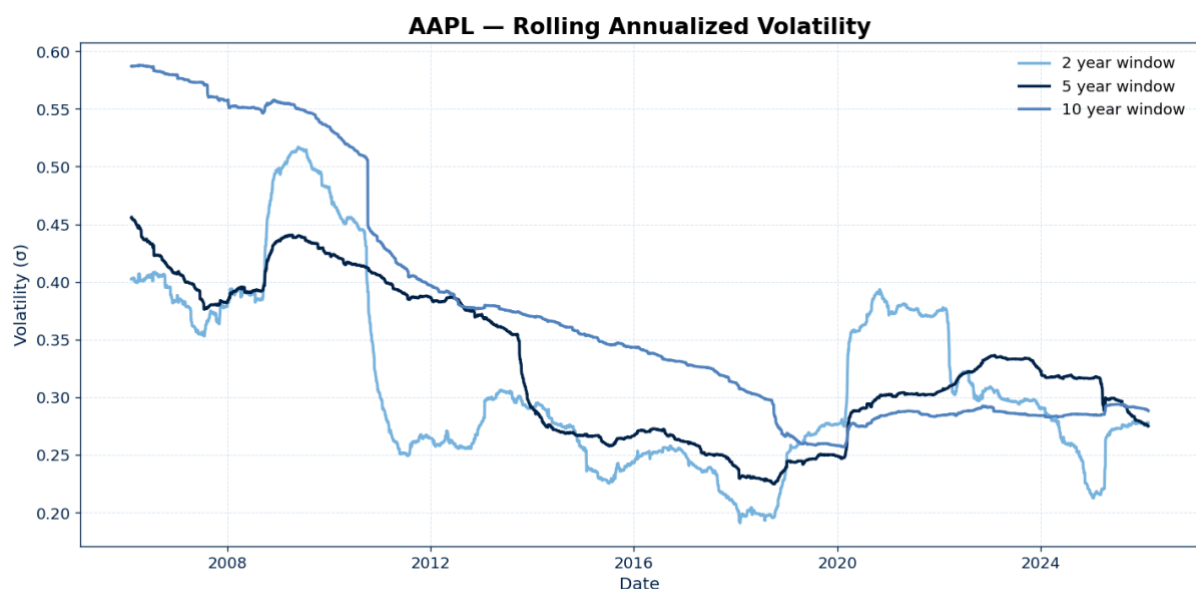
3)

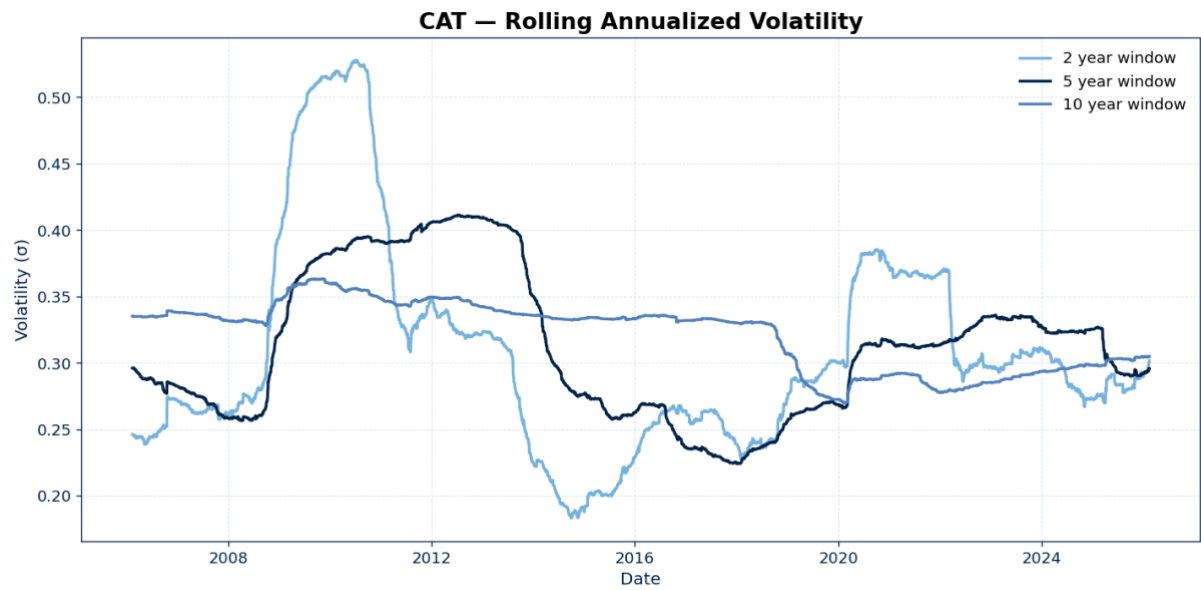
We have used **matplotlib** to generate the following plots, which are rolling parameter estimates of the mean and vol terms of both stocks for {2, 5, 10} year windows, generated using PX_LAST.

For the given two stocks, the parameters of mean and vol, are estimated using a rolling window of 2, 5, 10 years, of trading days, which we assume to be 252. The assumption that μ and σ are constant as stated in the Q, means that within the window of $[t - 1, t]$, we treated the GBM drift and vol terms as a constant, and “fit” the two values. Unweighted fitting, means that every daily return, received equal importance, which is not as methodically sound, as previous value importance decays. We computed daily log returns from PX_last, and we sample the mean and stdev and annualised using the 252-trading day conventions.

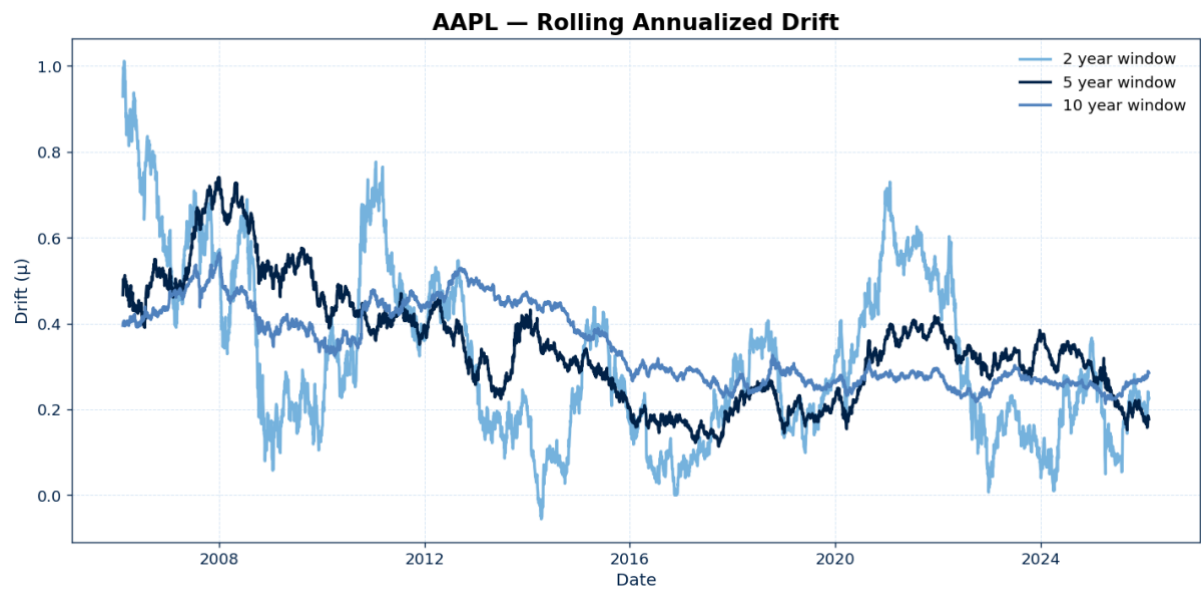
Notably, the **computed parameters are comparably**, because we are purely estimating the return parameters from the log returns, which are scale free. It doesn't not matter for the price level or dollars invested. So even if AAPL and CAT prices, and by extension, the dollar value of a certain position can change immensely across 20 years, the estimated drift and volatility are still comparable across the time, because it articulates the distribution of % changes, not the distribution of dollar terms.

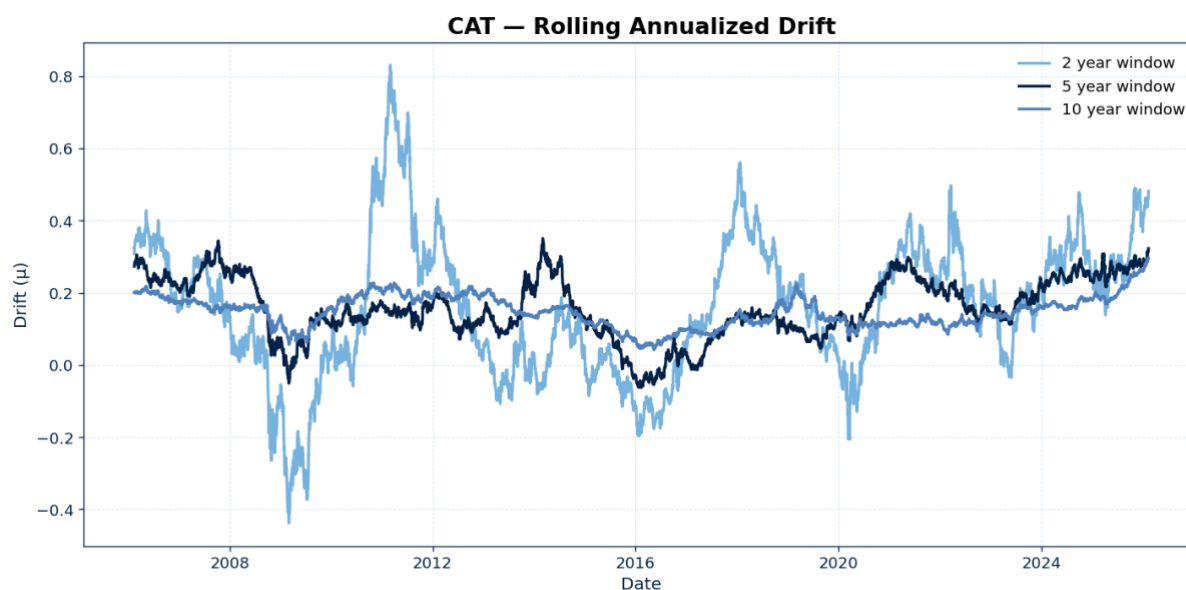
On an overall basis, the volatility term looks reasonably stable in the long-term but is clearly dependent on regimes. Drift is much less stable. The general trend, as expected is that the shorter windows, especially 2 years, are far less stable, whereas 10 years is the most stable.





For instance, looking at the volatility terms across the two equities, we can see that the 2Y window is extremely jumpy at stress periods, like during the GFC, and during the beginning stages of COVID, before quickly mean-reverting. 5Y tends to be a bit smoother. 10Y window, in this case, does seem to move slowly, and does capture long-run volatility. Pronounced spikes around major stress events, as mentioned prior, and followed by normalisation, does indicate generic volatility clustering.





On the other hand, the drift term is extremely noisy, and fluctuates widely, indicating that overall, the short-horizon drift estimation is not reliable. The 5Y drift is also quite variable, but the 10Y drift is the only one that looks somewhat stable, and differs across the different market regimes, like the recovery from 2008, but is probably extremely noisy, and has many confoundment factors, since it also covers the dotcom bubble burst.

Interestingly enough, volatility is stable enough for some interpretation, especially 5Y/10Y, but clearly affected by regimes. Drift is extremely noisy at the 2y marks, so for unweighted windows, it is worth comparing different l sizes.

4)

4. Exponential weighting

Some references say to use exponential weighting with $\lambda = 0.94$ or $\lambda = 0.97$ to calibrate to the last few weeks or about a month of data, respectively. These λ s correspond roughly to these periods because the weight becomes $1/2$ after approximately 14 days and 22 days, respectively.

To see the impact of exponential weighting with these weights, repeat the previous problem using exponential weighting, with $\lambda = 0.94$, and $\lambda = 0.97$.

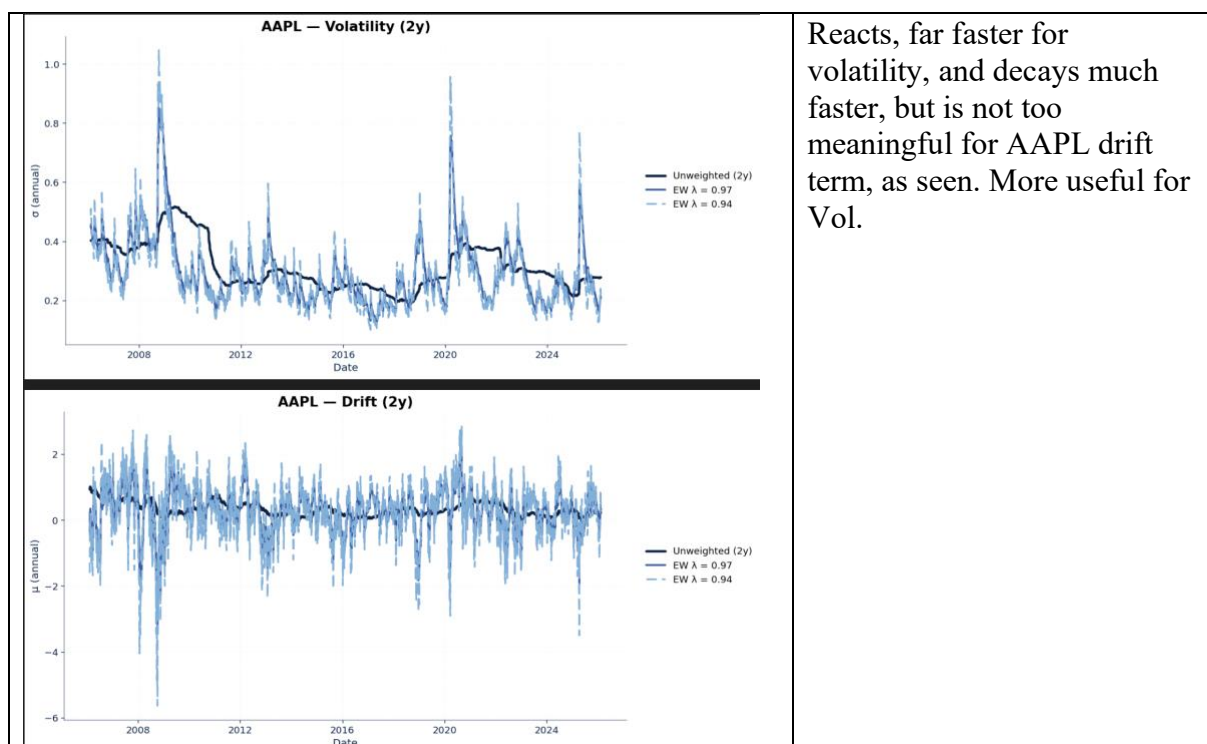
In the Q3 with unweighted fitting, each date t and window length l , GBM was estimated using the last l years, with each observation equal weight, but in Q4, the same rolling estimation is completed.

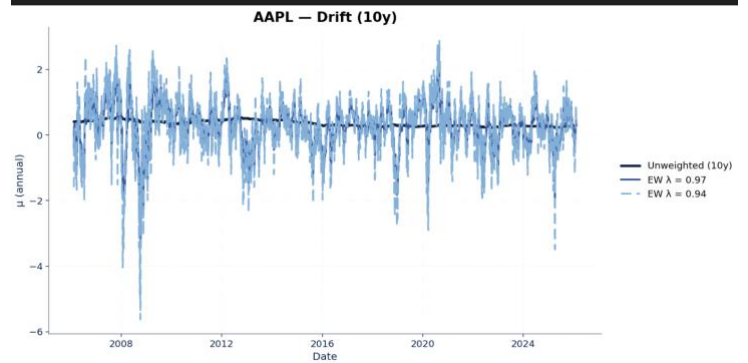
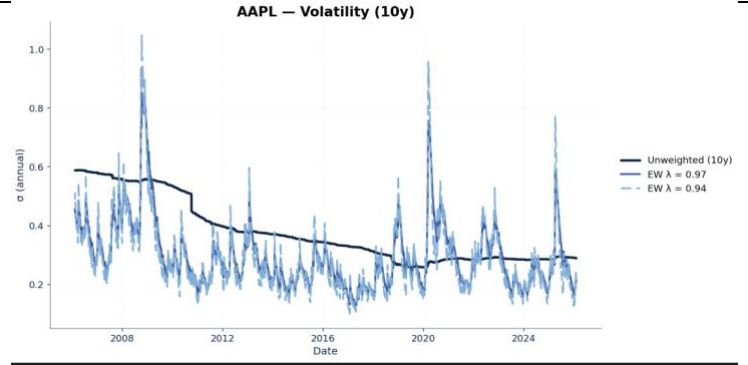
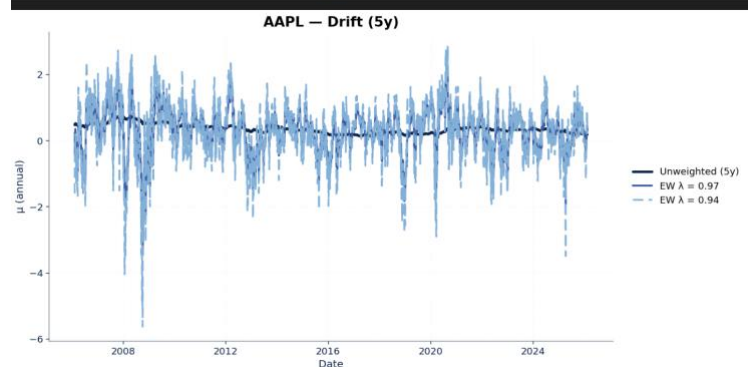
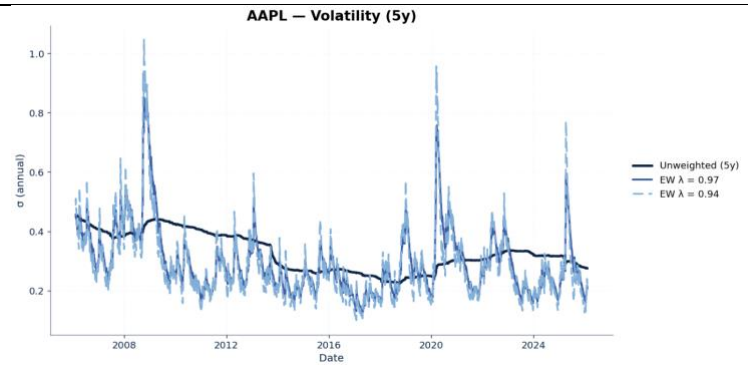
Notably, with the trailing window ending at weight are assigned such that decay is geometrically backward in time, such that the most recent returns has the largest weight, and a return k days old, has the weight proportional to $(1 - \lambda)\lambda^k$. Notably, a smaller λ decays

faster, so it places more emphasis on the last few weeks, and as such, it means that it is noisier. Vice versa for larger λ .

As volatility is driven by squared returns, if the market has a recent shock, exponential weighing makes those recent weightings dominate the returns. This means there are faster vol spikes in periods of stresses, and faster rise & decay. We expect this to be the case, for $\lambda = 0.94$ to swing more sharply than 0.97.

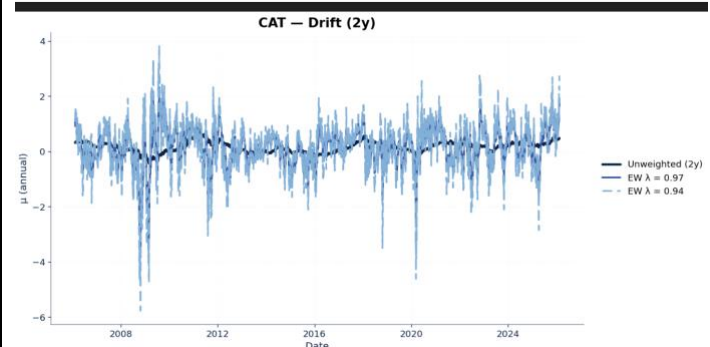
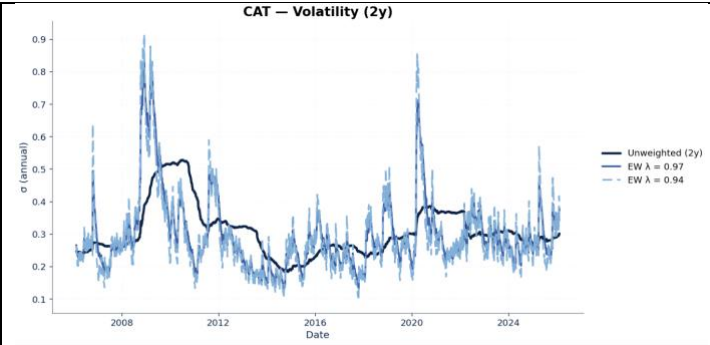
In addressing the parts, the estimated parameters are comparable over time because they are computed from log-returns, which are scale-free % returns. Drift and vol describe the distribution of proportional returns, not dollar levels. So, even if the AAPL and CAT prices differ substantively in the time, the estimated parameters remain directly comparable. Exponential doesn't change anything. In terms of stability, exponential weighting is materially changed, because recent observations, receive larger weights, vol responds more quickly to shocks and also decays more rapidly once the shock passes, with sharper spikes. However, this responsiveness, is at the cost of higher variability, with vol becoming more reactive, with drift becoming even noisier under exponential weighting, as such there is a trade-off.



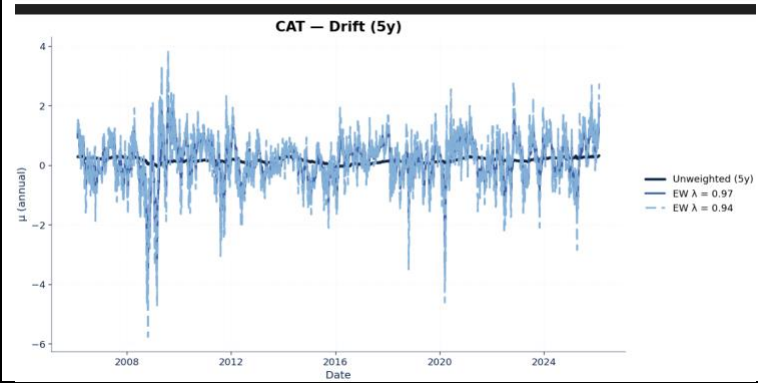
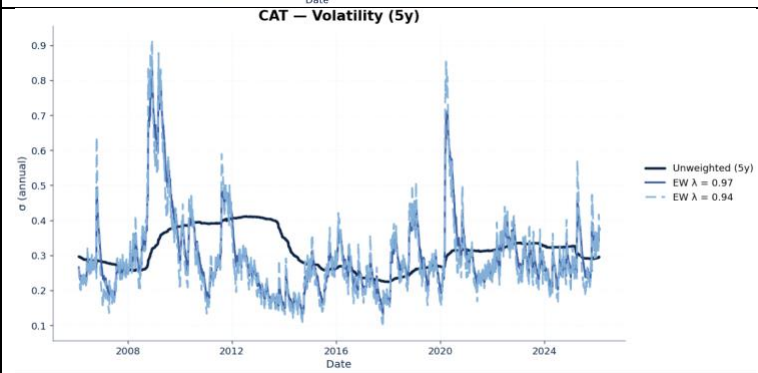


Likewise, more useful for volatility term.

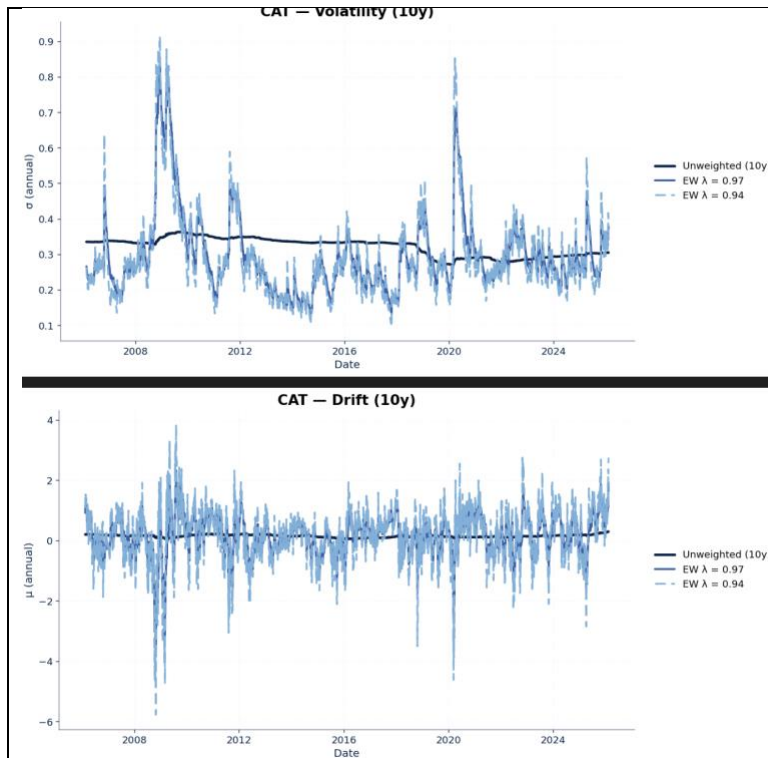
Now the 10Y volatility, is far more valuable, as it actually registers a volatility change in 2008 and 2020, whereas for unweighted, this barely registered. Drift term is now slightly more meaningful than before, as it is a little less noisy.



Like before, Vol more meaningful.

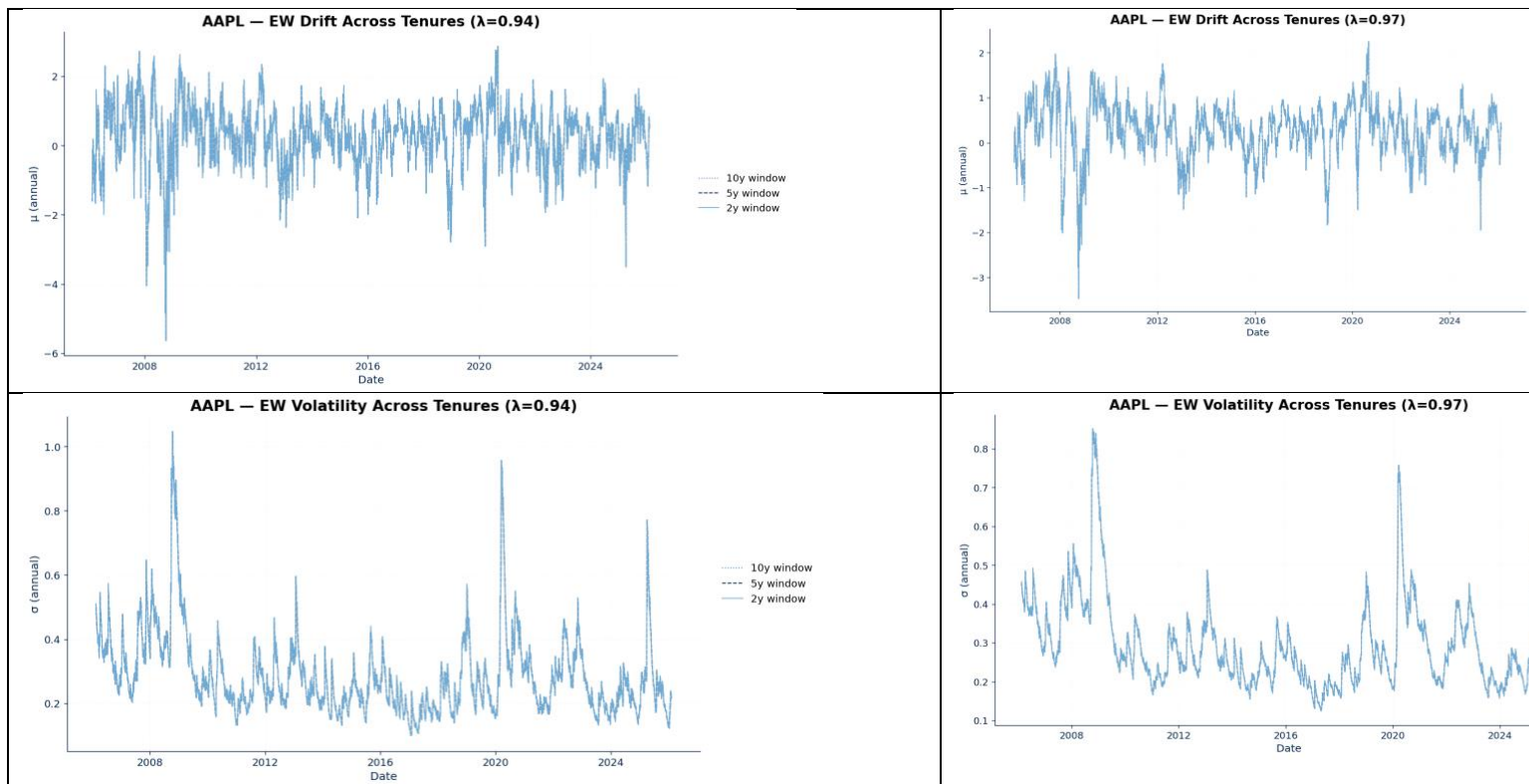


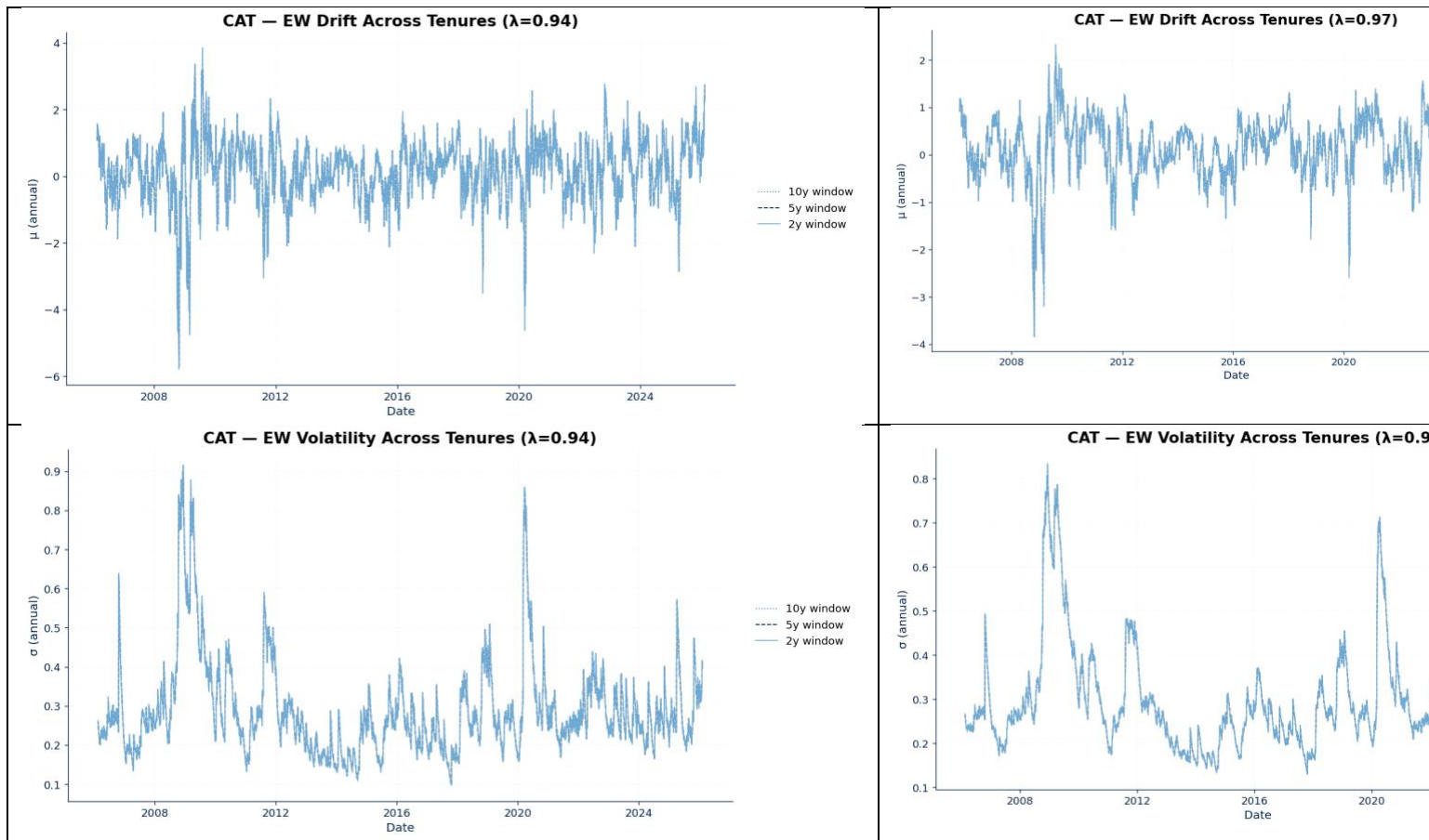
Like before, Vol more meaningful.



Like before, Vol more meaningful. But now, we can see that CAT drift term does have some interpretive meaning.

On the other hand, plotting each lambda, with its associated equity, showing the differences across each tenure, we have:





Notably, with exponential weighting, the effective lookback is set by λ , not by the window length, and the idea is that with lambda, anything older than the last few days is effectively redundant.

```

=== AAPL EW  $\lambda=0.94$  ===
Columns present: ['mu_2y', 'sigma_2y', 'mu_5y', 'sigma_5y', 'mu_10y', 'sigma_10y']
mu: 2y vs 5y -> max|diff|=1.834088e-13, mean|diff|=2.817110e-14
mu: 2y vs 10y -> max|diff|=1.834088e-13, mean|diff|=2.817110e-14
mu: 5y vs 10y -> max|diff|=0.000000e+00, mean|diff|=0.000000e+00
sigma: 2y vs 5y -> max|diff|=7.729928e-14, mean|diff|=4.105315e-15
sigma: 2y vs 10y -> max|diff|=7.729928e-14, mean|diff|=4.105315e-15
sigma: 5y vs 10y -> max|diff|=0.000000e+00, mean|diff|=0.000000e+00

=== AAPL EW  $\lambda=0.97$  ===
Columns present: ['mu_2y', 'sigma_2y', 'mu_5y', 'sigma_5y', 'mu_10y', 'sigma_10y']
mu: 2y vs 5y -> max|diff|=9.044854e-07, mean|diff|=1.517795e-07
mu: 2y vs 10y -> max|diff|=9.044854e-07, mean|diff|=1.517795e-07
mu: 5y vs 10y -> max|diff|=4.440892e-16, mean|diff|=1.025249e-17
sigma: 2y vs 5y -> max|diff|=3.373052e-07, mean|diff|=2.715610e-08
sigma: 2y vs 10y -> max|diff|=3.373052e-07, mean|diff|=2.715610e-08
sigma: 5y vs 10y -> max|diff|=1.110223e-16, mean|diff|=3.552890e-18

=== CAT EW  $\lambda=0.94$  ===
Columns present: ['mu_2y', 'sigma_2y', 'mu_5y', 'sigma_5y', 'mu_10y', 'sigma_10y']
mu: 2y vs 5y -> max|diff|=1.811884e-13, mean|diff|=2.848043e-14
mu: 2y vs 10y -> max|diff|=1.811884e-13, mean|diff|=2.848043e-14
mu: 5y vs 10y -> max|diff|=0.000000e+00, mean|diff|=0.000000e+00
sigma: 2y vs 5y -> max|diff|=5.107026e-14, mean|diff|=4.045619e-15
sigma: 2y vs 10y -> max|diff|=5.107026e-14, mean|diff|=4.045619e-15
sigma: 5y vs 10y -> max|diff|=0.000000e+00, mean|diff|=0.000000e+00

=== CAT EW  $\lambda=0.97$  ===
Columns present: ['mu_2y', 'sigma_2y', 'mu_5y', 'sigma_5y', 'mu_10y', 'sigma_10y']
mu: 2y vs 5y -> max|diff|=9.327257e-07, mean|diff|=1.539812e-07
mu: 2y vs 10y -> max|diff|=9.327257e-07, mean|diff|=1.539812e-07
mu: 5y vs 10y -> max|diff|=4.440892e-16, mean|diff|=7.940213e-18
sigma: 2y vs 5y -> max|diff|=2.894247e-07, mean|diff|=2.661723e-08
sigma: 2y vs 10y -> max|diff|=2.894247e-07, mean|diff|=2.661723e-08
sigma: 5y vs 10y -> max|diff|=1.110223e-16, mean|diff|=3.552890e-18

```

As a primer, there is a fundamental trade-off for more timely parameter estimates, but trades off for more variability or noise, especially in the drift term, as we can see above. Sometimes, that also applies to volatility.